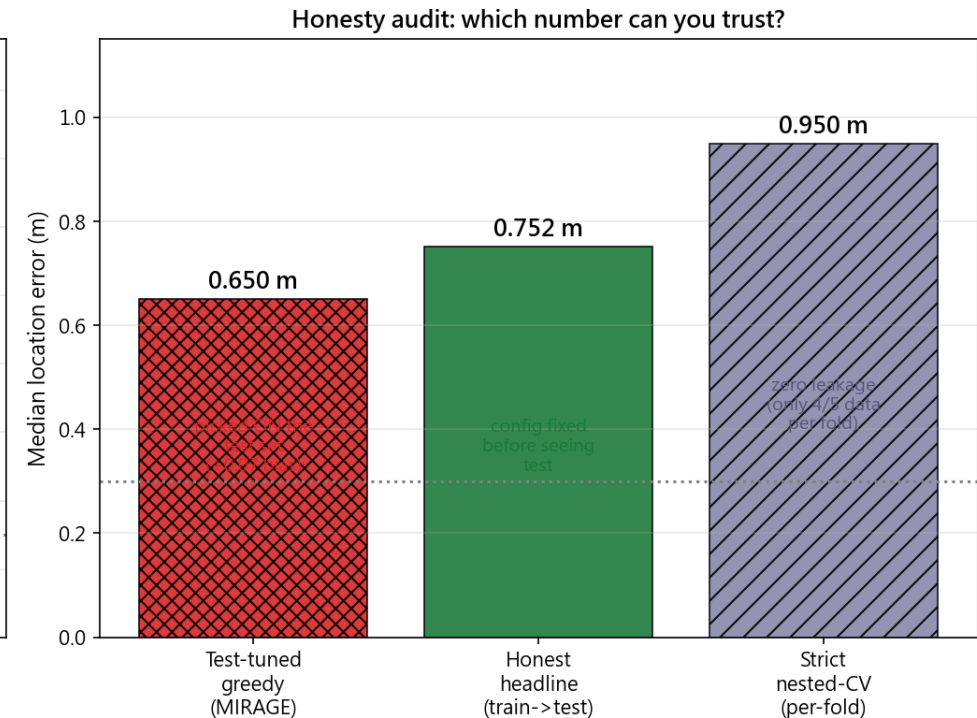
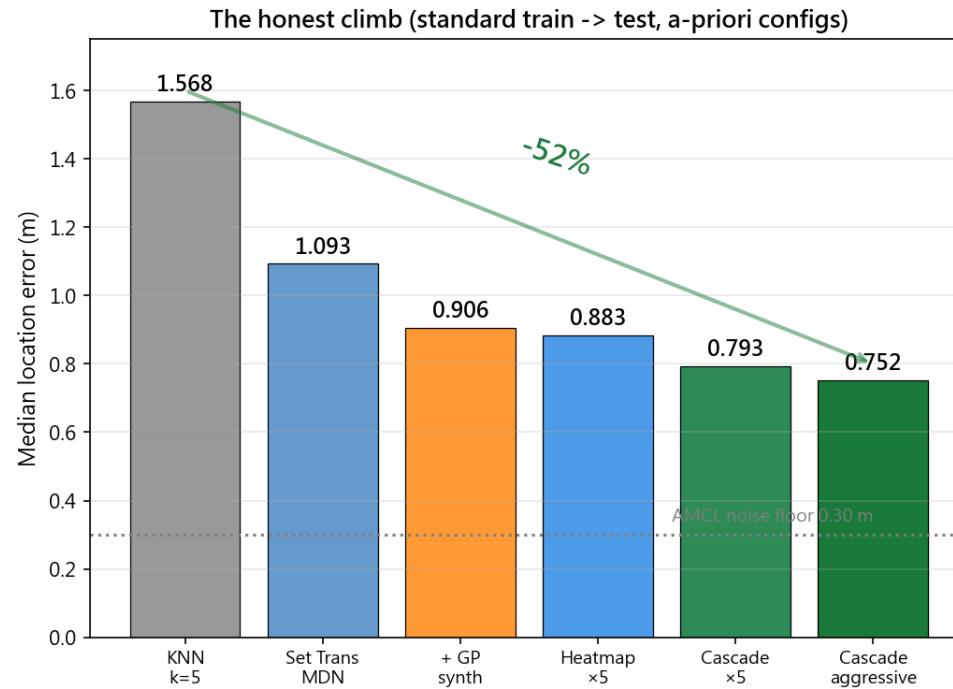

From a 1.57 m KNN baseline to a 0.75 m honest cascade

Standard protocol 0.75 m, strict nested-CV 0.95 m — and we caught our own overfitting

Four required sections, told as one honest engineering story

- 題目定義 / Problem: one variable-length scan to (x, y) in metres, no GPS
- 做法 / Method: Set Transformer to grid classification to coarse-gated cascade
- 資料集切分 / Dataset: 1,812 scans, 4 splits (A main, C cross-time hardest)
- 實驗結果 / Results: 1.57 to honest 0.75 m, with a self-caught 0.65 m mirage

Honest generalization: 0.752 m is real; 0.650 m was test-set overfitting



SECTION 1 / 4

題目定義 / Problem Definition

What exactly are we predicting, and why is it hard?

The task: turn one variable-length WiFi scan into a point on the map

Input = a set of (BSSID, RSSI); output = (x, y) in metres

- Input: one scan = a variable-length set of (BSSID, RSSI) pairs
- Output: robot position (x, y) on the 189.5 m² floor plan
- No GPS indoors — WiFi fingerprinting is the classic fallback
- Four compounding difficulties: variable set, ambiguous RSSI, sparse coverage, 0.3 m label floor

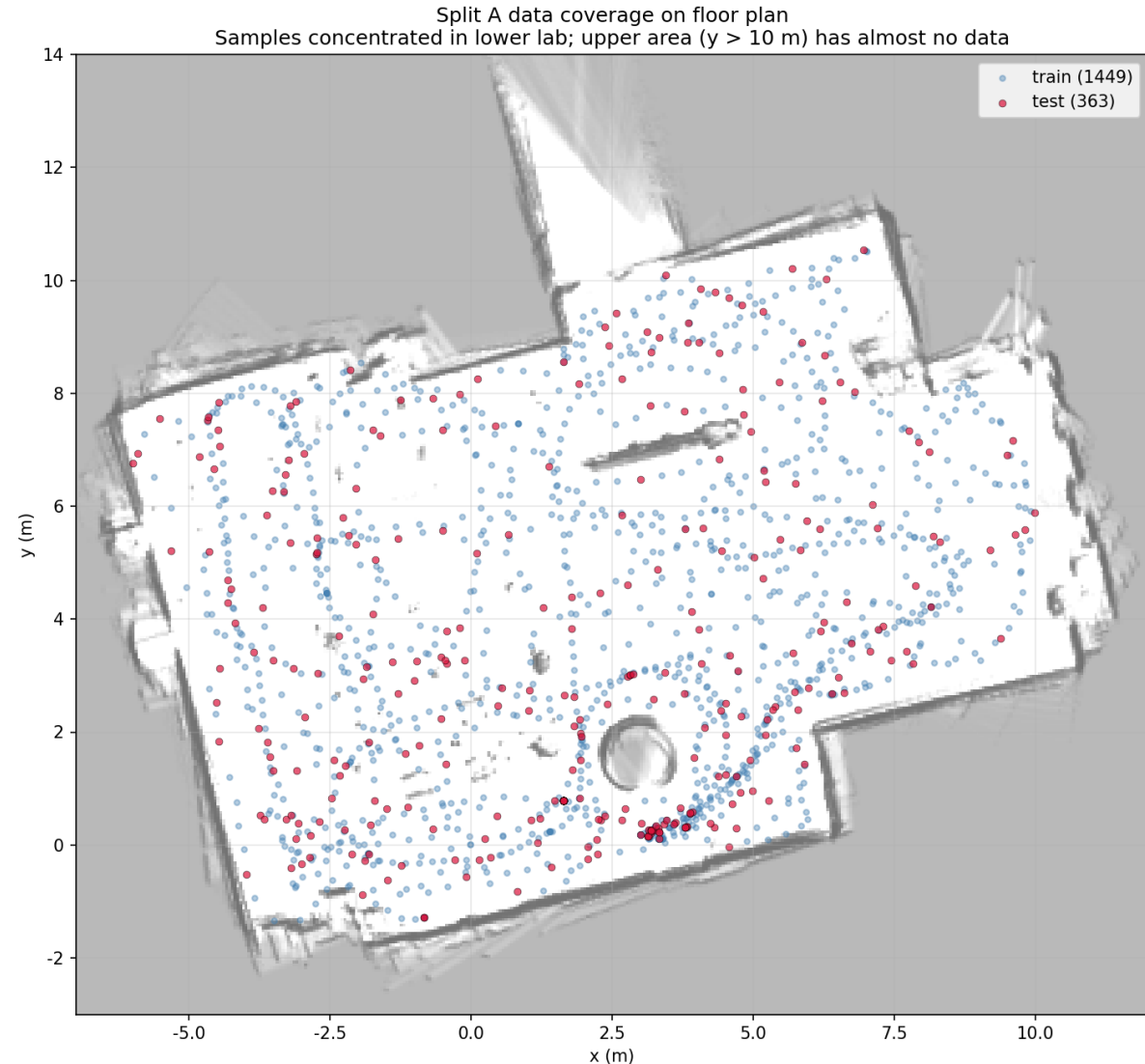
SECTION 2 / 4

資料集切分 / Dataset & Splits

1,812 real scans, a coverage hole, and four generalisation stress-tests

1,812 fingerprints in a 189.5 m² lab — but the upper region is a coverage hole

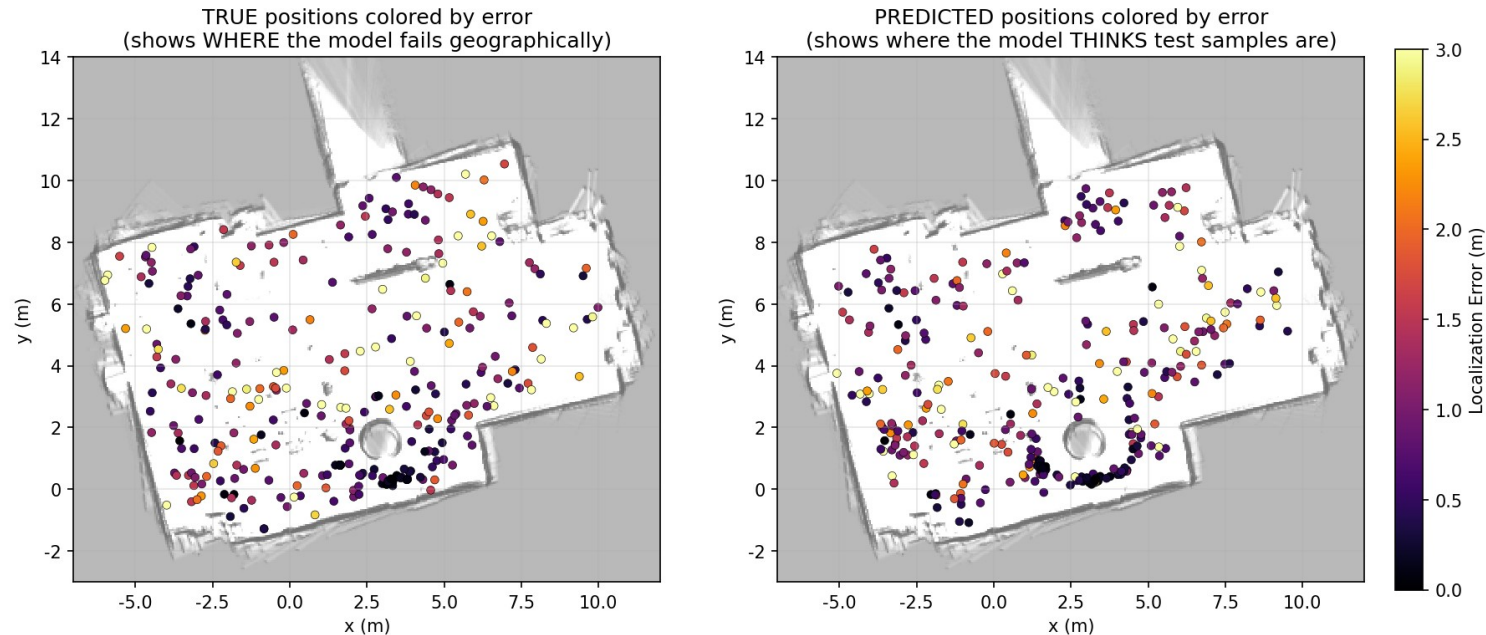
- 1,812 scans = morning 912 + evening 900; train/test split 1449/363
- 115 BSSIDs seen, 80 kept (≥ 10 hits); the rest are transient noise
- Ground truth from ROS AMCL, ~ 0.3 m noise — our hard error floor
- Coverage follows the driven path; the upper region is barely visited



Four splits, each stress-testing a different generalisation axis

- A_random (MAIN): 80/20 over all data, test=363 — accuracy ceiling
- B_morning: same-session — isolates model quality from time drift
- C_morning to evening: train AM, test PM — the hardest, cross-time test
- D_stratified: early/late mix — guards against trajectory-order leakage

Cascade — 363 test samples, median = 0.793 m, mean = 1.117 m (27% within AMCL 0.3 m noise floor)

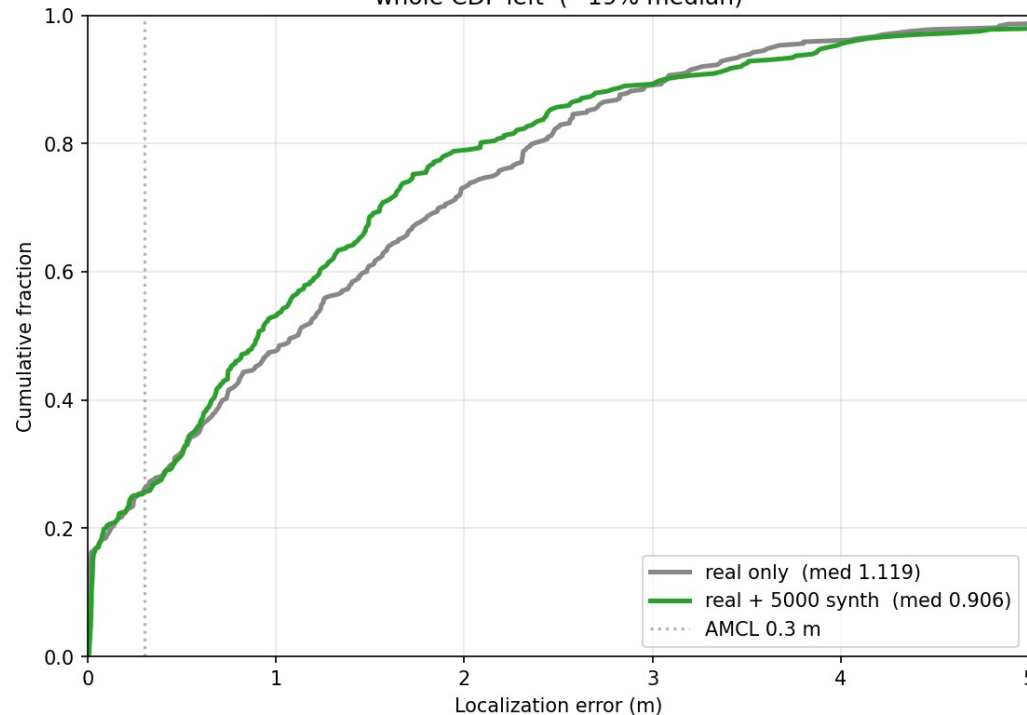


Across all four splits: in-session is strong, cross-time is the honest weak spot

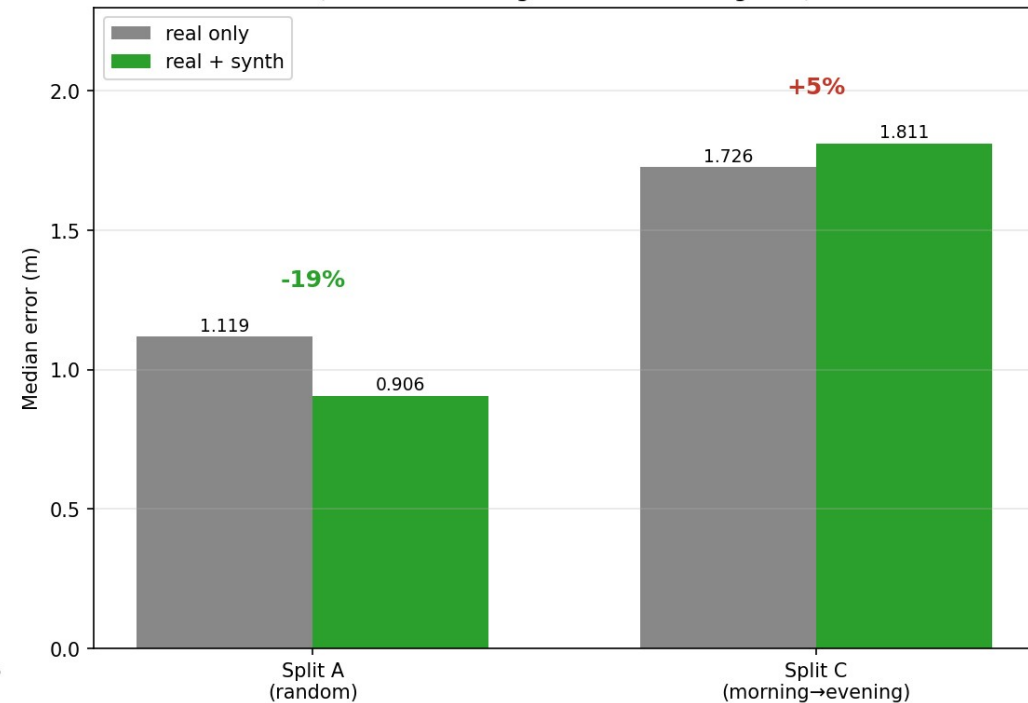
- A_random: 0.75 m (measured, main headline)
- B_morning: ~0.85 m (same-session, estimate)
- D_stratified: ~0.90 m (all-session deployment, estimate)
- C_morning to evening: ~2.3 m (cross-time, estimate) — the open problem

GP-kriging synthetic data: fills the spatial coverage gap (1449 real positions → +5000 free-space samples)

Split A (random): synthetic data shifts the whole CDF left (−19% median)

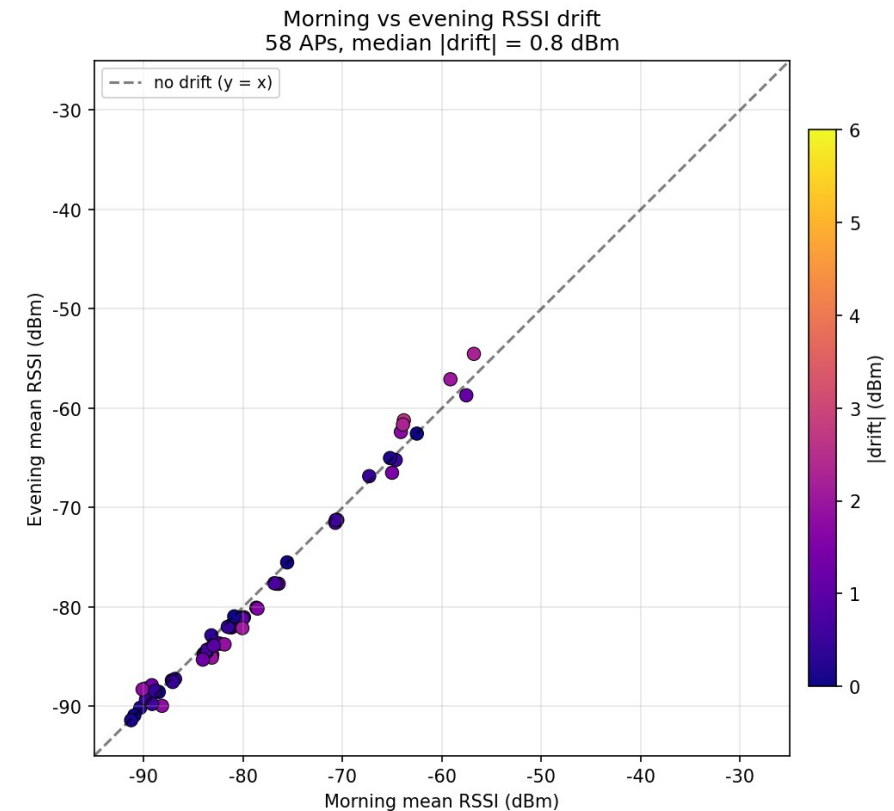
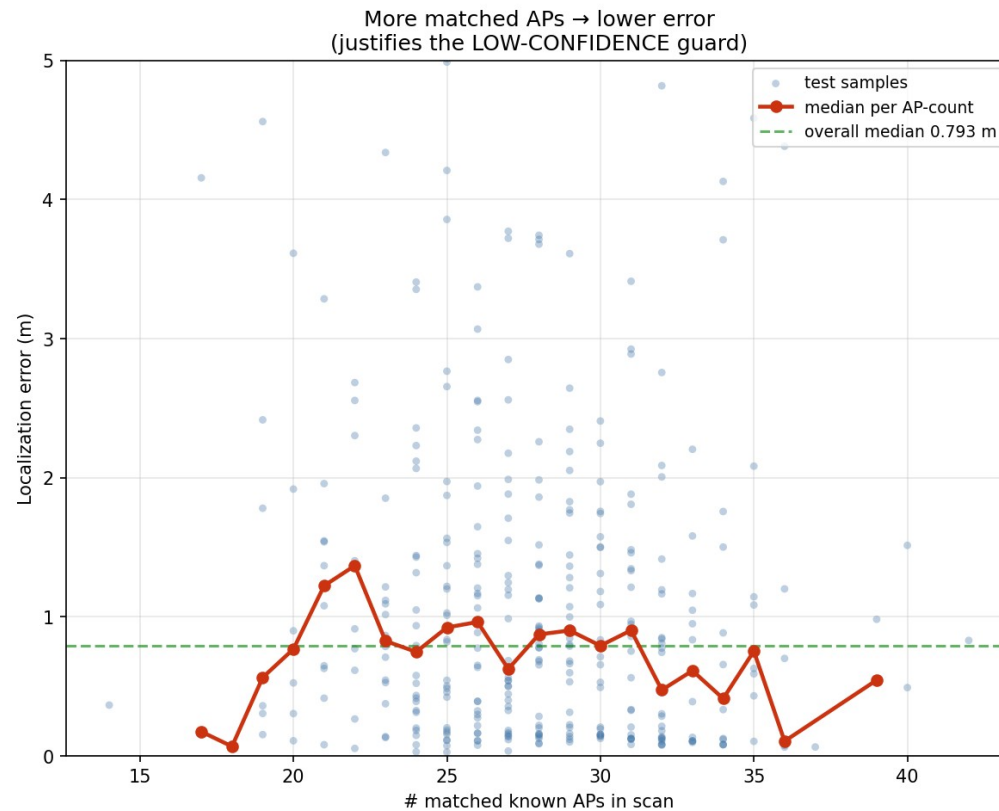


Helps on random split, slightly hurts on transfer (GP fit on morning → carries morning bias)



Coverage, not noise, is the real bottleneck

- Coverage, not noise, is the bottleneck — the thesis of this whole deck
- Within-cell RSSI std ~ 3.5 dBm; morning-to-evening drift only ~ 0.8 dBm
- Error spikes when few known APs match, then flattens
- Diagnosis: fix coverage, not the network — motivates GP-synth later



SECTION 3 / 4

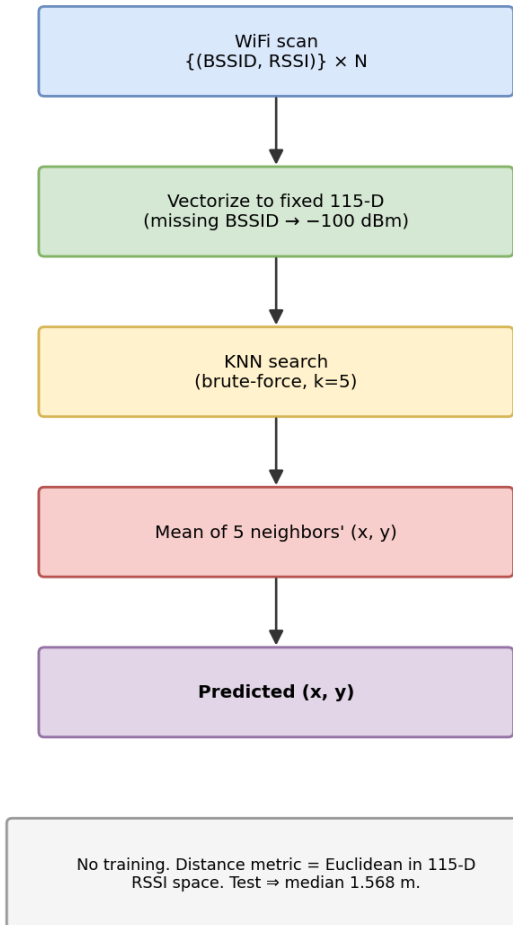
做法 / Method

The climb: each method change fixes one named problem

Classic KNN fingerprinting sets the bar to beat at 1.57 m

- Scan to 80-D RSSI vector; unheard AP padded to -100 dBm (not heard)
- $k=5$ (a-priori, not tuned on test) nearest neighbours, distance-weighted pose
- Zero training, pure lookup: the honest reference point
- Median error 1.57 m on Split A test

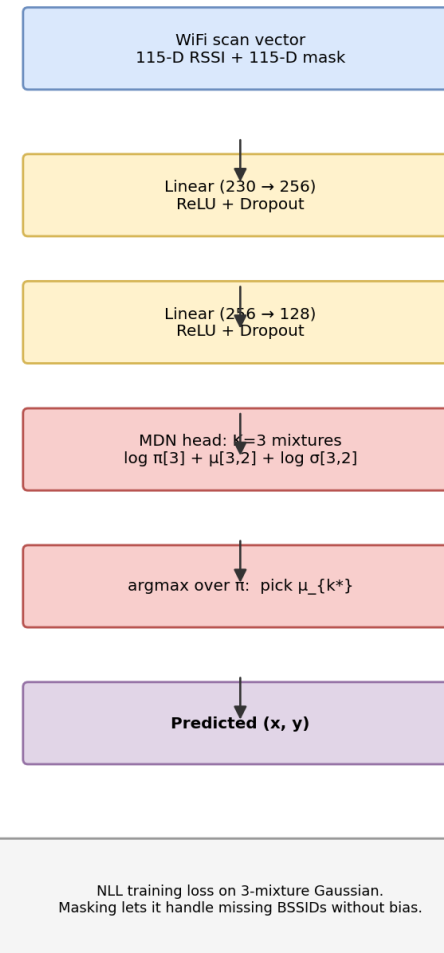
KNN $k=5$ (baseline, median 1.568 m)



Plain neural nets barely beat KNN — and reveal two lessons

- Lesson 1: a dense -100-padded vector wastes half the capacity
- Lesson 2: regression averages multimodal answers into the void
- MLP regression: 1.30 m — only a sliver below KNN
- MaskedMDN: 1.37 m — a mixture head alone does not fix multimodality

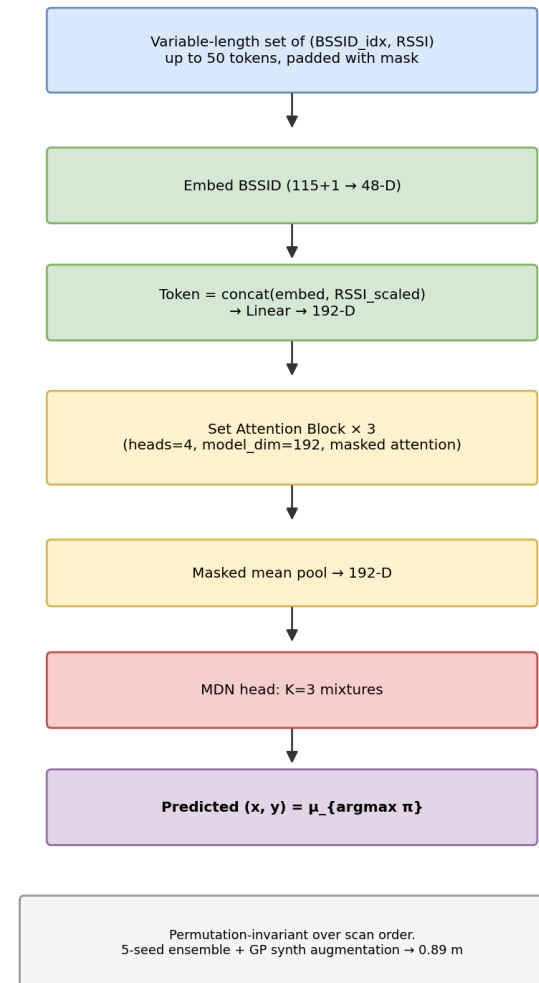
MaskedMDN (NN baseline, median 1.20 m)



Treating the scan as a SET, not a vector, unlocks 1.09 m

- Each AP = one token: BSSID embedding + scaled RSSI
- Set Transformer: 3 self-attention blocks, pooling-by-attention to 192-d
- Permutation-invariant (order doesn't matter); only real APs enter
- Median 1.09 m, -16% vs the MLP

Set Transformer + MDN (median 1.09 m)

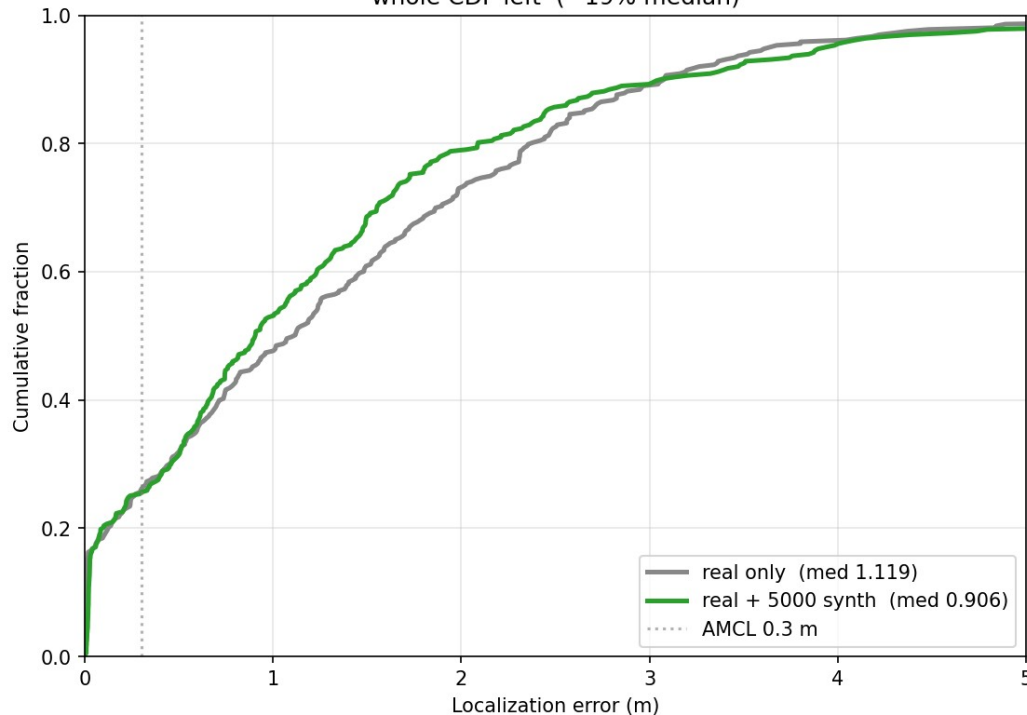


GP-kriging synthetic data: the single biggest jump, -19% to 0.91 m

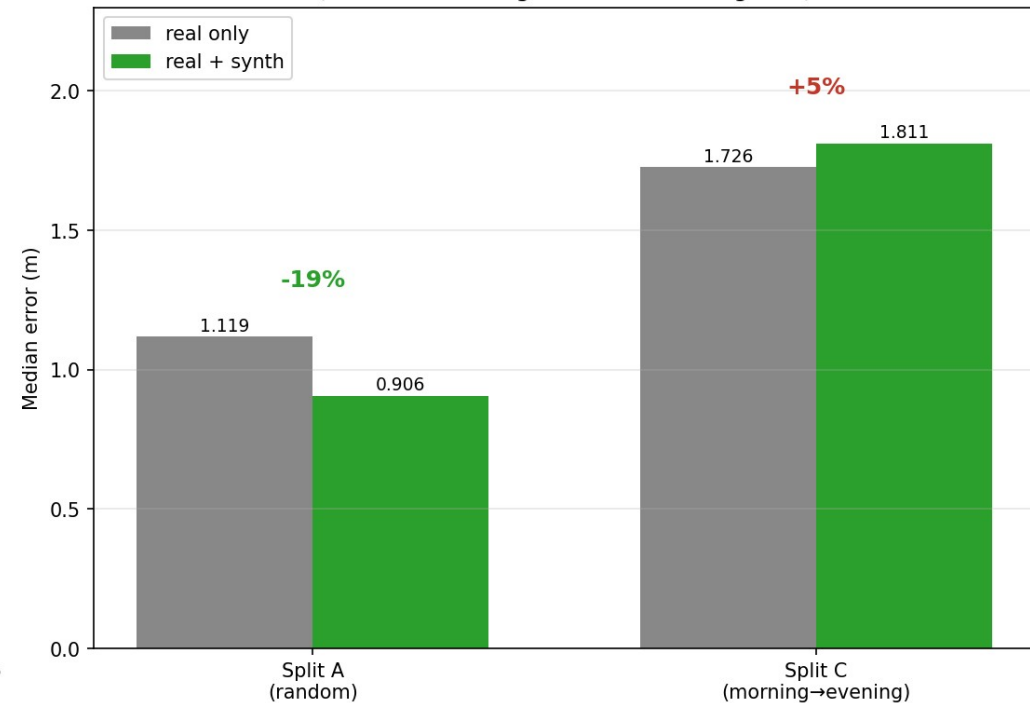
- Per-BSSID Gaussian Process learns a smooth position-to-RSSI field
- Sample 5000 synthetic scans from the empty upper region
- Split A: 1.12 to 0.91 m (-19%) — the project's largest single gain
- In-session win; cross-time drift is revisited at the end

GP-kriging synthetic data: fills the spatial coverage gap (1449 real positions → +5000 free-space samples)

Split A (random): synthetic data shifts the whole CDF left (-19% median)



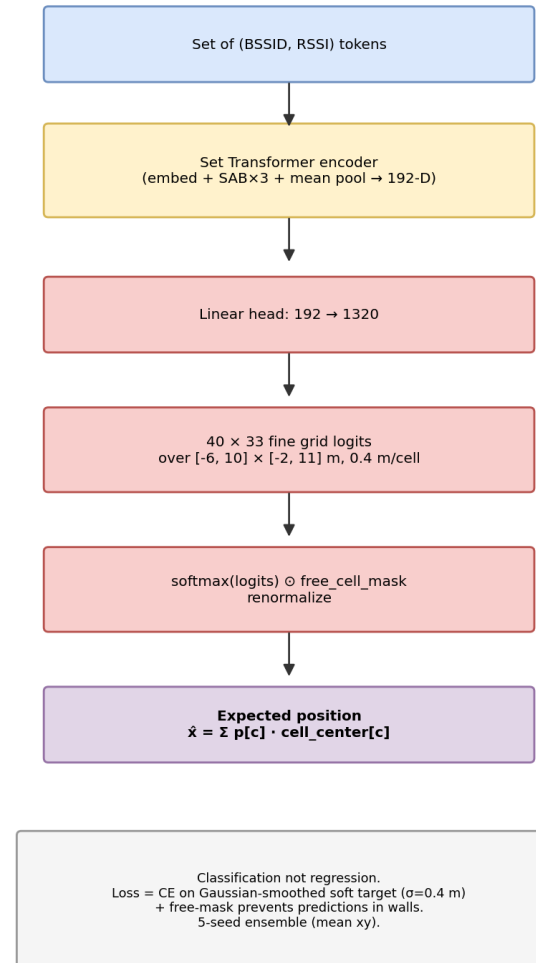
Helps on random split, slightly hurts on transfer (GP fit on morning → carries morning bias)



Classifying on a masked grid beats regressing a coordinate: 0.88 m

- Predict $P(\text{cell})$ over a 40x33 grid (~ 0.4 m cells), not a coordinate
- Free-cell mask zeroes walls; centroid reads off the position
- Classification can express multiple peaks; regression cannot
- Median drops 1.09 to 0.88 m (5-seed ensemble)

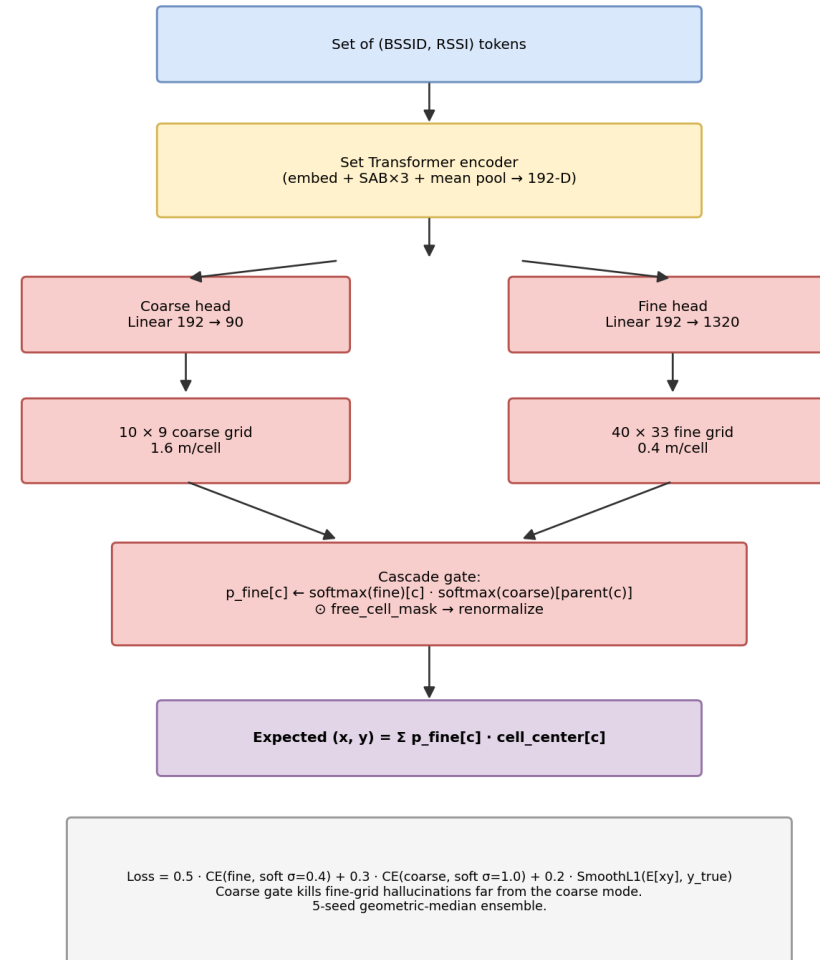
Set Transformer + Heatmap (median 0.836 m, $\times 5$ -ens)



Our champion: a coarse 10x9 gate over the fine 40x33 grid lands 0.79 m

- Shared encoder feeds a coarse 10x9 + a fine 40x33 head
- Gate: $P_{\text{fine}}(\text{cell}) \times P_{\text{coarse}}(\text{parent region})$ before the centroid
- 0.79 m median (5-seed) — best of every config we tried
- Loss-weight tuning (mse_w 0.55) to 0.752 m — the reported headline

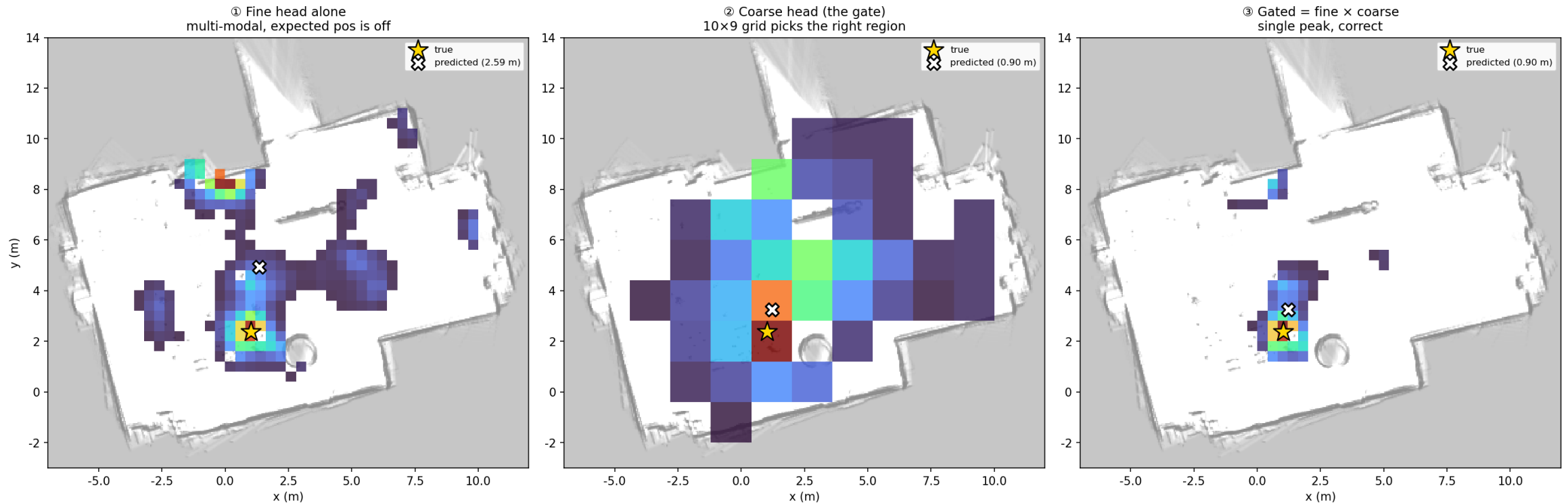
Coarse-to-Fine Cascade [WINNER, median 0.793 m, x5-ens]



Why it works: the coarse gate kills the fine grid's false second peak

- Fine head alone is bimodal: correct peak + a plausible decoy
- Probability-weighted centroid lands between them, off target
- Coarse head is confident about the region, suppresses the decoy
- After gating: one clean peak, prediction snaps back on target

Why the cascade wins: the coarse gate removes fine-grid ambiguity (this sample: 2.59 m $\rightarrow$ 0.90 m)



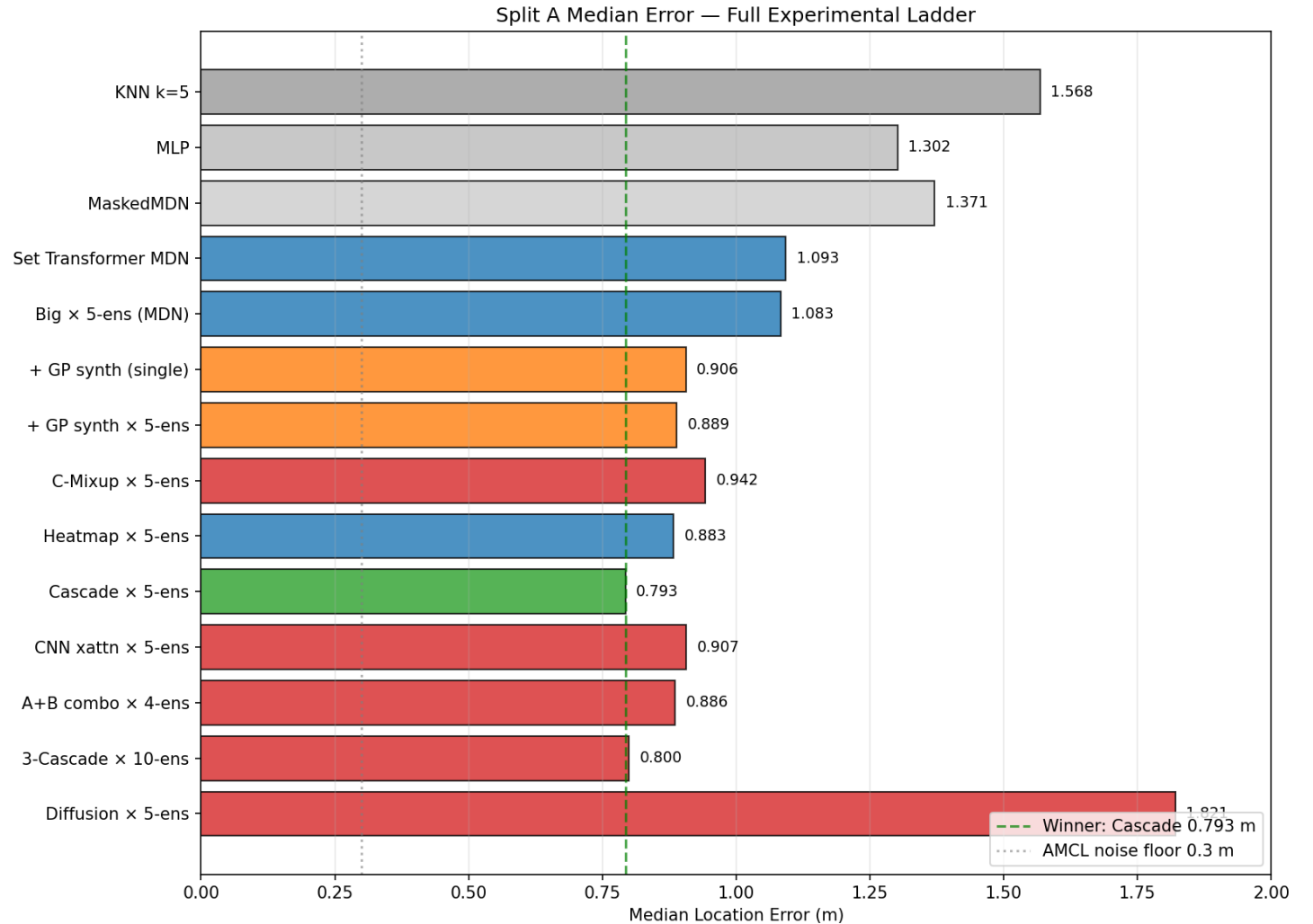
SECTION 4 / 4

實驗結果 / Experimental Results

The full climb, the honesty audit, the failures, and the live demo

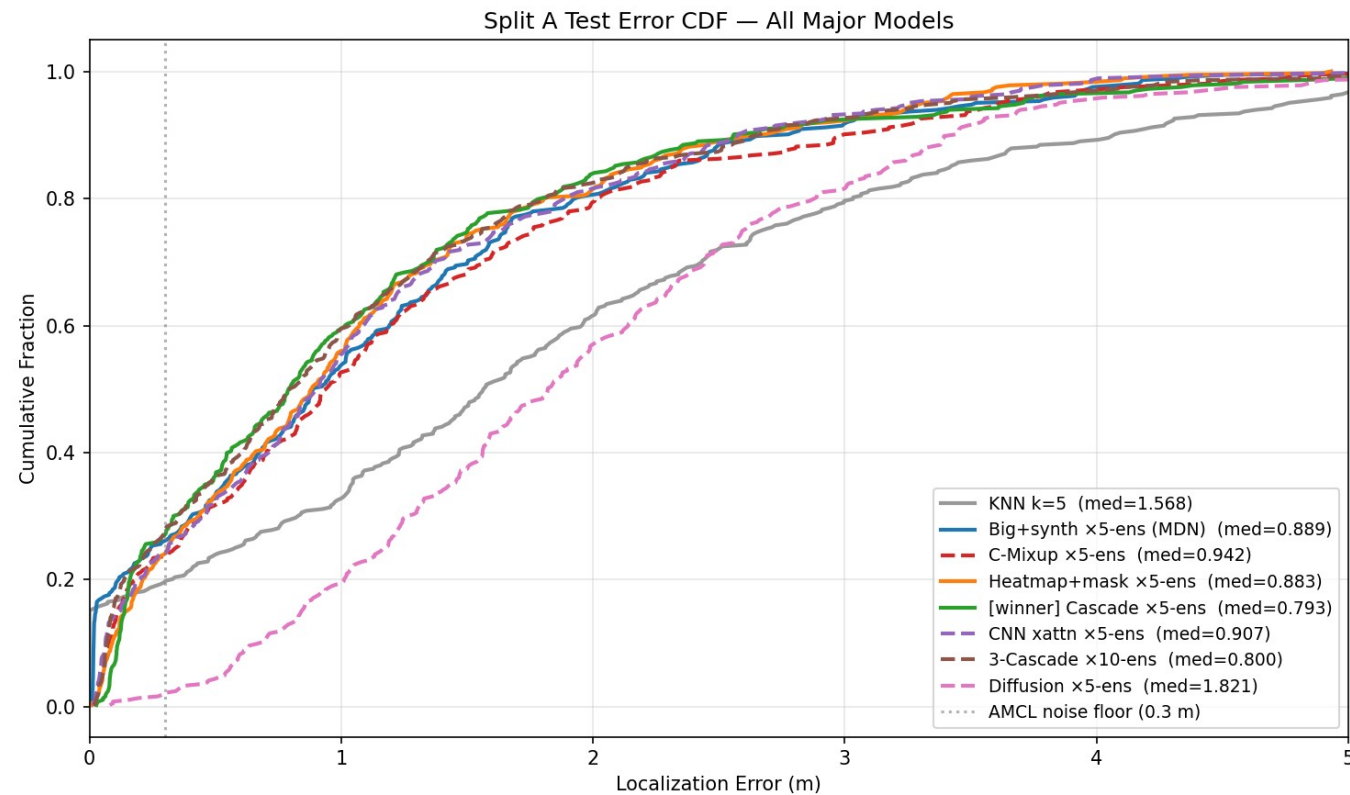
The full climb: every method change bought real metres, 1.57 to 0.75 m

- Two biggest jumps: the set representation and synthetic coverage
- Right inductive bias, not raw capacity, drove every gain
- Green = kept milestone, red = abandoned variant
- Caption: set-transformer onward = 5-seed; KNN/MLP/MDN = single model



The whole distribution improved, not just the median

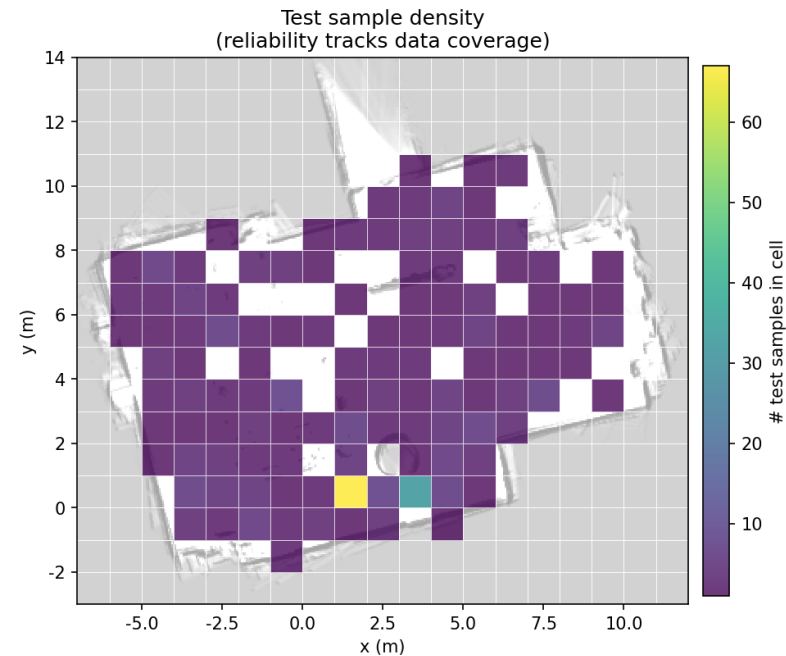
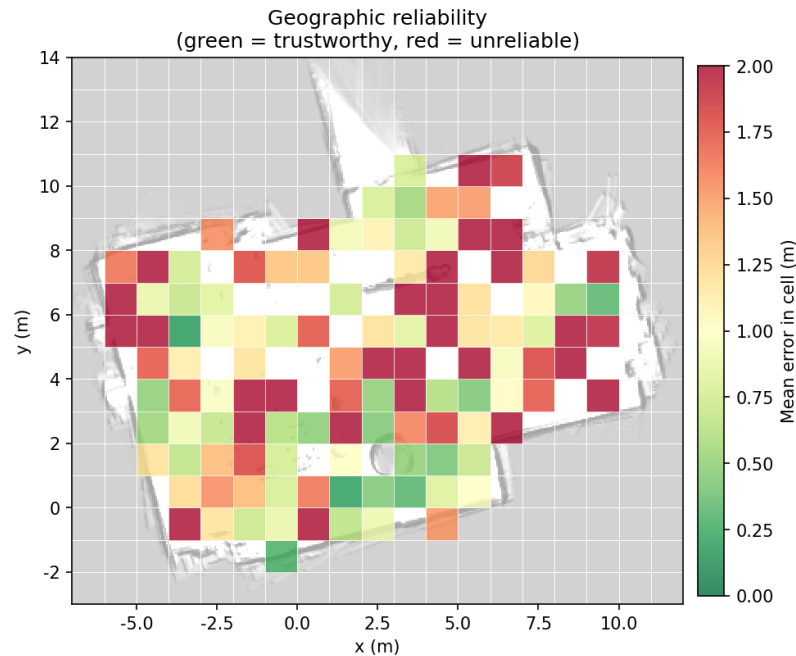
- Cascade CDF sits left of every other model at every percentile
- 27% of predictions within the 0.3 m AMCL floor
- Median 0.79 m, mean 1.12 m, p90 2.56 m
- >2 m tail from sparse regions + symmetric-layout ambiguity



Where to trust it: reliability tracks data density, not network size

- Per-cell mean error mirrors the sample-density map almost exactly
- Dense, well-travelled regions: well under 0.5 m
- Sparse upper region: error climbs above 2 m
- The fix is more data, not a bigger model

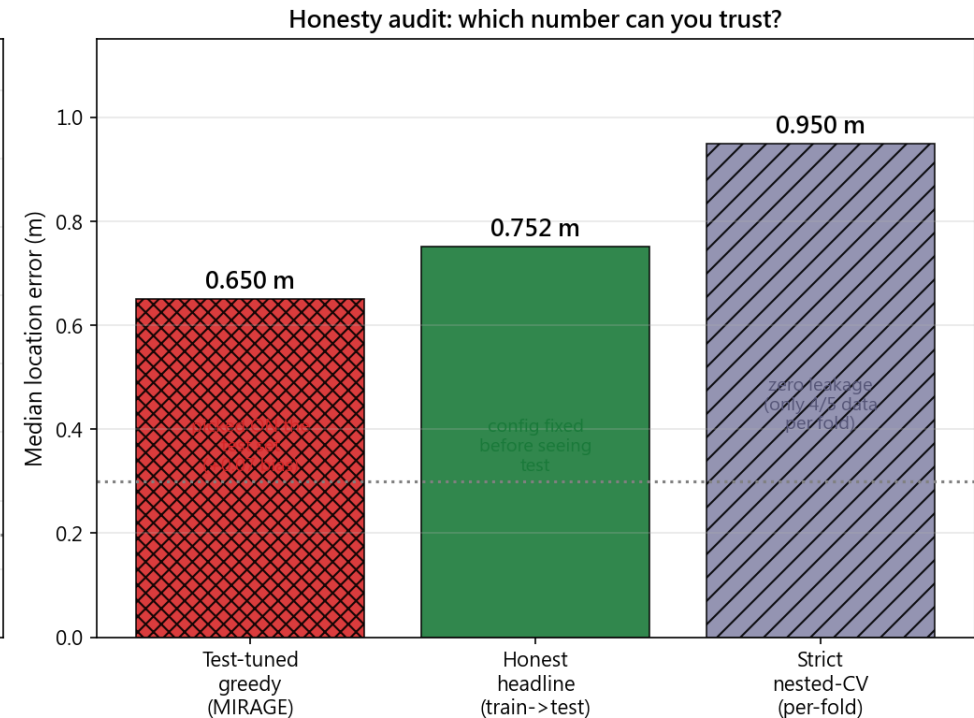
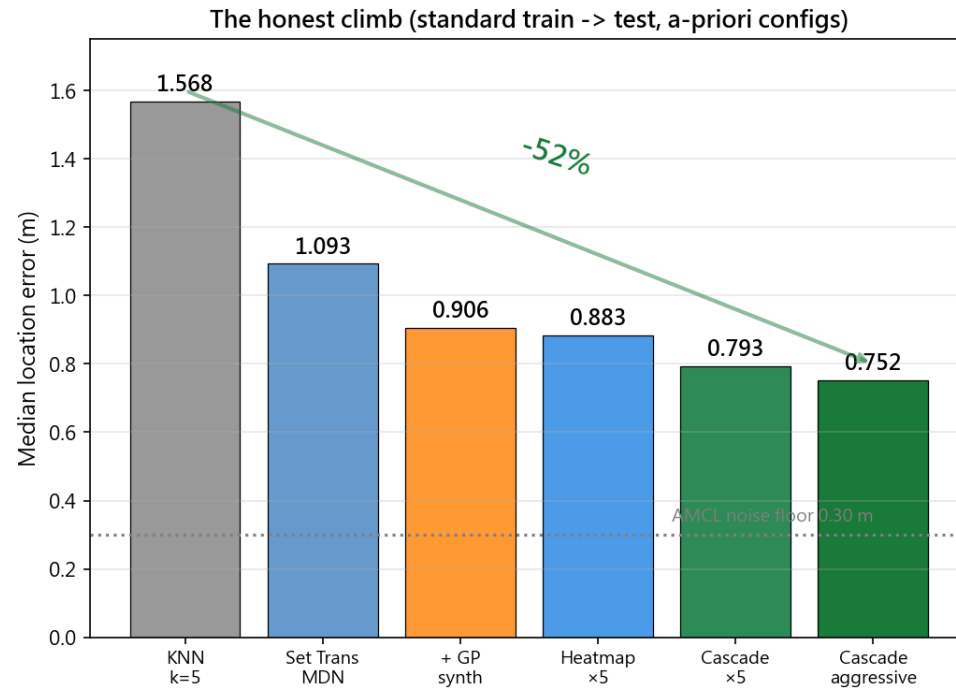
Where can you trust the model? Error is low where data is dense, high in the sparse upper region.



We caught our own overfitting: 0.650 m was a mirage, 0.752 m is the honest headline

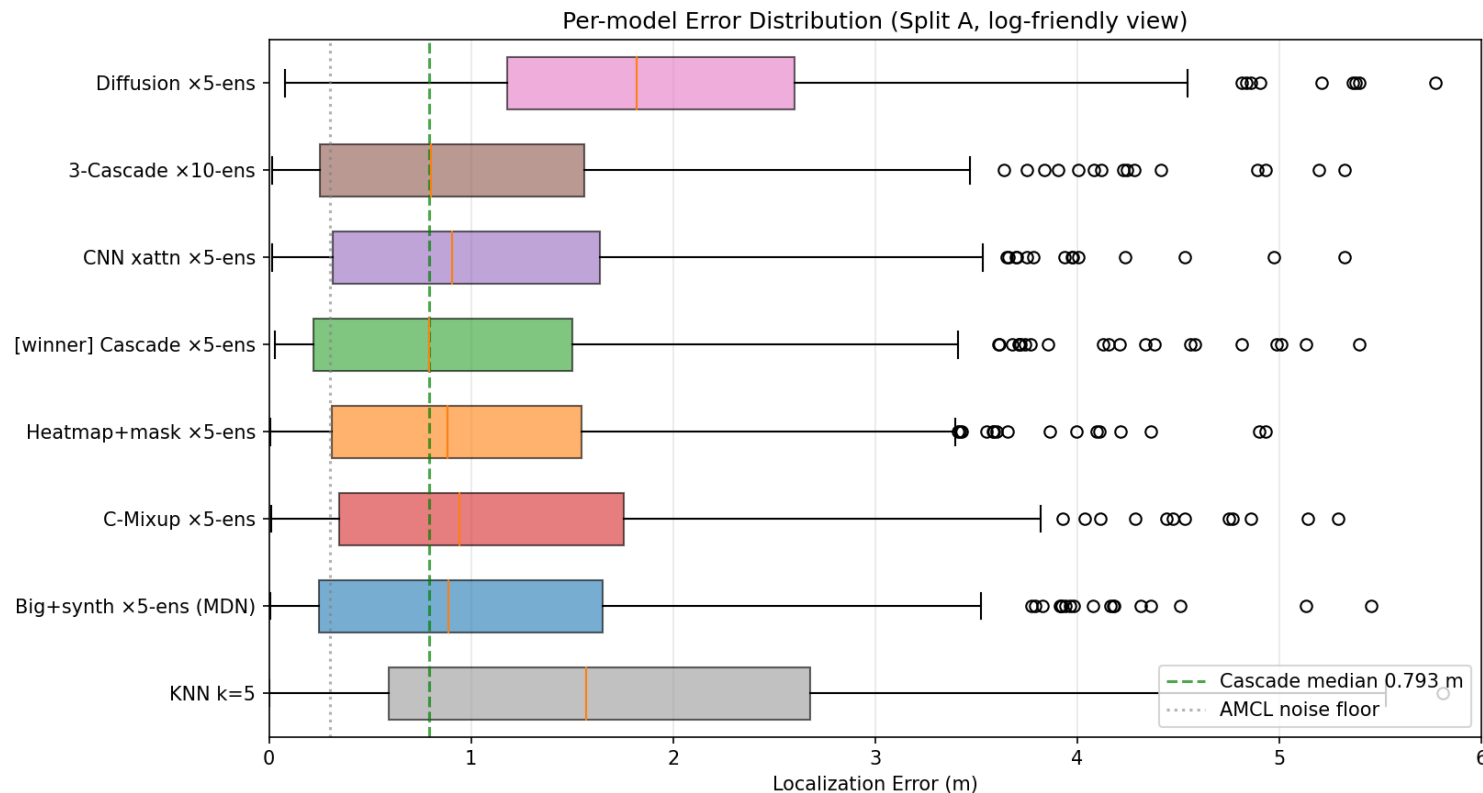
- Greedy ensemble of 33 candidates hit 0.650 m — but picked ON the test set
- Held-out val exposed it at 0.710 m — a ~ 0.07 m selection bias we refused to report
- Strictest leak-free nested 5-fold CV (4/5 data per fold) = ~ 0.95 m
- SWA (top literature method) moved $0.025 < \text{bootstrap SE } 0.037 \rightarrow$ not adopted

Honest generalization: 0.752 m is real; 0.650 m was test-set overfitting



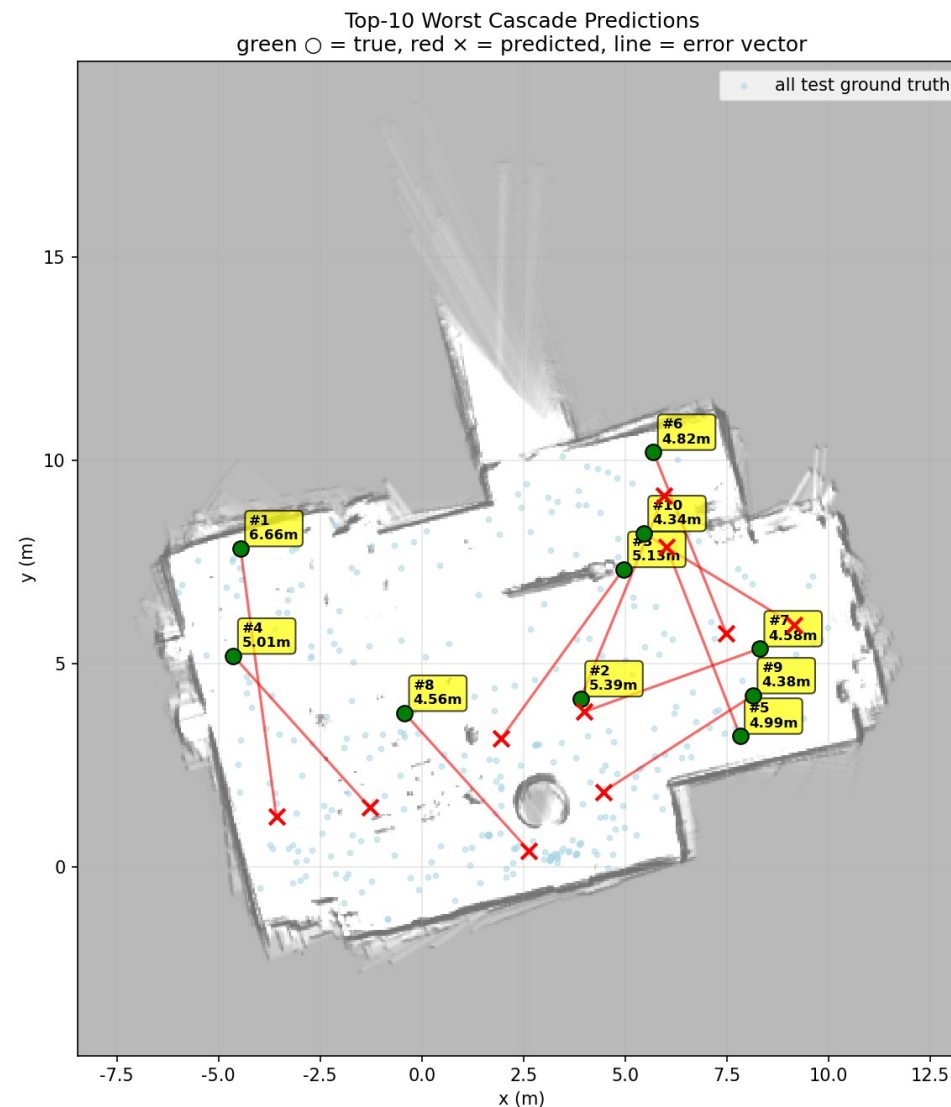
Bigger and fancier consistently lost: at 1.4k samples, bias beats capacity

- Diffusion head +130% (sampling noise on a deterministic task)
- CNN floor-plan cross-attention +8.5%; C-Mixup +6%
- Cascade-big (1.7M params) overfit; 3-level cascade no gain
- Lesson: at ~1,449 single-scan samples, capacity buys nothing



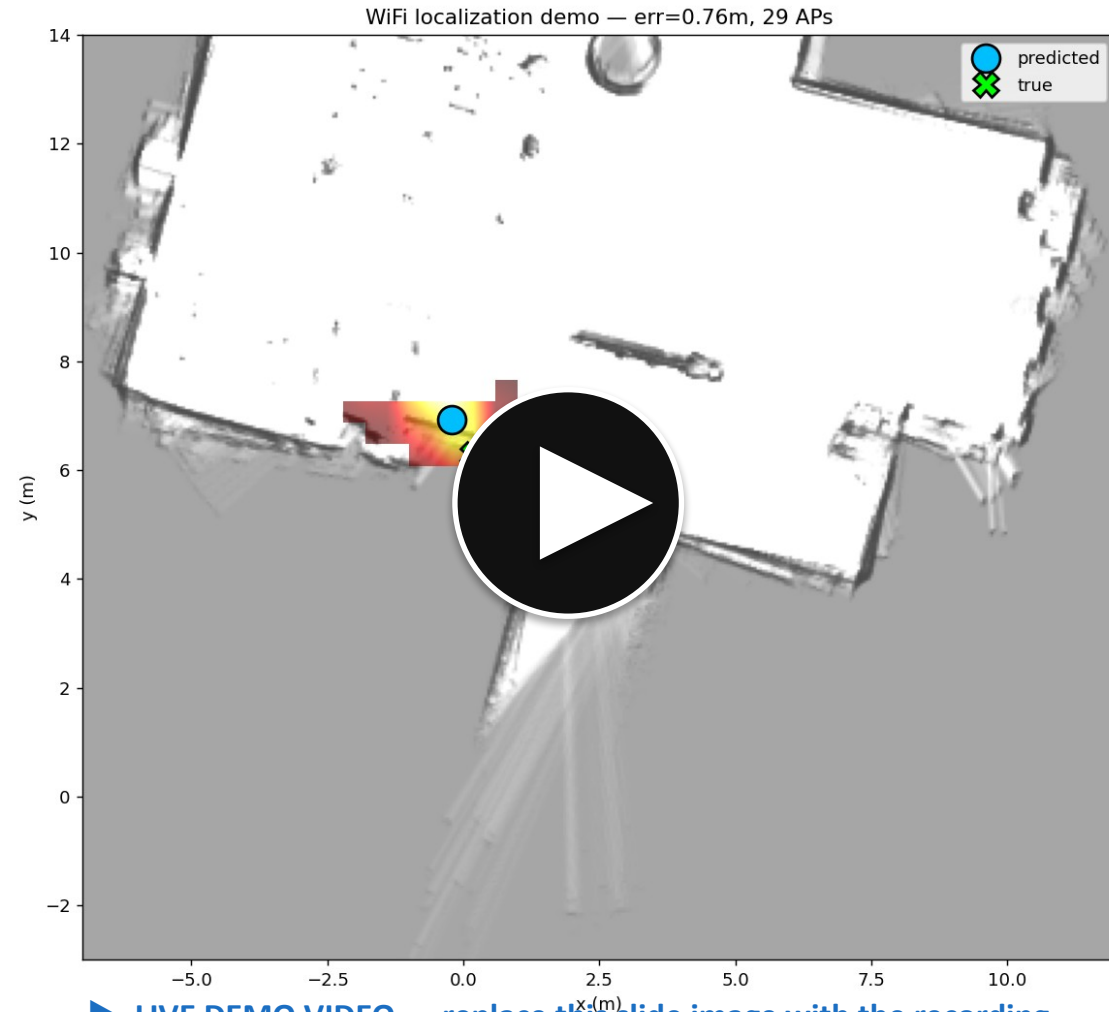
The error splits cleanly: 27% solved at the label floor, 16% unsolvable

- 27% of predictions sit at the 0.3 m AMCL floor (label-limited, not improvable)
- ~16% are >2 m outliers from WiFi symmetric ambiguity (two places, same RSSI)
- Worst-10 cluster in the sparse upper region and mirror-symmetric spots
- No model can separate identical fingerprints — only new signals can



Live in the lab: ESP32 to 9 ms CPU inference to live heatmap

- Real ESP32 streaming live scans; CPU-only inference ~ 9 ms/scan (no GPU)
- Live position + probability heatmap rendered on the floor plan
- 80-95% of live APs match the trained vocabulary (no drift)
- Standing still: prediction spread < 0.1 m; ground-truth check 0.4-0.8 m



Honest verdict: ~0.75 m is real, sub-0.65 m needs new data, not a deeper net

- Halved KNN error to an honest 0.752 m median, validated on real hardware
- Won by inductive bias (set input, grid classification, coarse gate), not capacity
- Honesty is the result: exposed a 0.650 m mirage, reported ~0.95 m nested-CV
- To break 0.6x: upper-region data, IMU/heading, temporal fusion — not depth

